

Strategic Contract Offering and Channel Choices in Live Streaming E-Commerce

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Abstract—The emergence of live streaming has revolutionized retail, altering how e-tailers engage with their customers. However, incorporating a live streaming channel may intensify channel competition and profit-sharing losses, potentially eroding the e-tailer's overall profitability. A streamer faces a tradeoff in contract provision: it must balance maximizing its profits with expanding its market. This article investigates a streamer's contract offering decisions and an e-tailer's channel selection strategy. We analyze two types of contracts that a streamer can offer—agency and reselling contracts—and three channel strategies for an e-tailer: not adopting live streaming, self-streaming, and third-party streaming. Our findings reveal that the streamer offers contracts aligned with its profit preferences only when the network effect is strong and the hassle cost is low. Otherwise, the streamer may deviate from its preferred contract and choose a suboptimal strategy. From the e-tailer's perspective, no single strategy dominates all scenarios. The e-tailer should avoid opening a live-streaming channel when the hassle cost is very high. If the network effect is low and the hassle cost is moderately high, the e-tailer should choose self-streaming; otherwise, it should opt for third-party streaming. Interestingly, our research also shows that neither the e-tailer nor the streamer necessarily has to lower the live streaming channel price. Furthermore, implementing live-streaming does not always cannibalize the traditional channel demand or lead to an expansion of total market demand. This paper provides practical guidance for both streamers and e-tailers in making strategic contract and channel decisions in the live-streaming retail landscape.

Managerial Relevance Statement—This study provides valuable managerial insights for managers and policymakers in the context of the increasing prevalence of live streaming. By constructing an interactive model that incorporates nuanced consumer behavior specific to live streaming, it uncovers optimal strategies for both streamer contract offerings and e-tailer channel selection. This paper enables streamers (managers) to understand: first, under what market conditions to offer which type of contract to best facilitate cooperation with e-tailers; and second, how to ensure they can choose optimal contracts aligned with their profit preferences, without sacrificing those preferences to secure collaboration.

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For e-tailers (managers), this study offers insights into channel selection decisions and pricing strategies across different channels. Furthermore, this study advises policymakers to avoid promoting live streaming as a one-size-fits-all solution, advocating instead for a supportive environment that fosters a variety of sustainable retail models. Policymakers can also play a key role in facilitating cooperation between e-tailers and streamers, for example, through policies that promote technological innovation such as AI-powered chatbots, which can reduce the hassle cost and improve the consumer experience. This article also contributes to SDG 8.

Index Terms—Channel selection, contract offering, live streaming, retailing.

I. INTRODUCTION

AS AN emerging retail format, live streaming e-commerce is significantly transforming the global retail landscape. By integrating social media, real-time video, and online retailing, it enhances the consumer shopping experience [1]. Compared with traditional retailing, live streaming introduces interactive features—such as real-time Q&A and on-screen comments—that enhance social interaction. Moreover, its immersive environment, characterized by live product demonstrations and time-limited promotions, amplifies consumers' purchase intentions [2], [3], [4]. China has witnessed rapid growth in live streaming. In 2023, the size of the Chinese live streaming e-commerce market reached USD 680 billion and is projected to grow to USD 1.15 trillion by 2026.¹ Meanwhile, Western markets are also accelerating their adoption of live streaming. For instance, YouTube has begun integrating live streaming shopping features, providing merchants with new opportunities for digital commerce.²

Traditional e-commerce primarily presents static product detail pages with text and images [5], which often limit consumers' ability to fully perceive product value and attributes [6]. To overcome this challenge, e-tailers are turning to self-streaming as a strategic alternative. For example, the Chinese apparel brand Peacebird leveraged self-streaming, achieving a monthly gross merchandise volume exceeding USD 156 million in 2021.³ Self-streaming grants e-tailers greater control over their sales channels and enhances consumers' perception of product value.

¹<https://www.statista.com/statistics/1276120/livestream-e-commerce-sales-united-states/> (accessed on December 2, 2024)

²<https://www.forbes.com/councils/forbestechcouncil/2025/02/03/how-live-shopping-is-changing-the-retail-landscape-across-the-globe/> (accessed on August 28, 2025)

³<https://www.forbes.com/sites/laurenhallanan/2021/06/21/sales-associate-live-streaming-has-become-a-must-have-for-brands-in-china/> (accessed on August 28, 2025)

TABLE I
REAL-WORLD EXAMPLES

Mode	Description
Traditional	<i>Christopher Ward</i> adopts a traditional mode, relying mainly on text and image displays to sell products through its official online store. ⁴
Self-streaming	<i>Apple</i> adopts a self-streaming mode, selling products not only through its online store but also via its official live-streaming channel. ⁵
Agency	<i>Fresh</i> adopts an agency mode, selling through its online channel while collaborating with celebrity streamer <u>Jiaqi Li</u> to promote and sell its products. ⁶
Reselling	<i>Nong Sao Food Co., Ltd.</i> adopts a reselling mode, selling products online and supplying goods to <u>Dongfang Zhenxuan</u> for live-streaming sales. ⁷

Note: “*Italics*” represent e-tailers, and “underlines” represent third-party streamers.

However, this approach also presents a dual challenge. On the supply side, the adoption of self-streaming may cannibalize demand from the traditional channel and impose additional operational expenses on e-tailers. On the demand side, consumers must spend time watching product demonstrations, potentially incurring the hassle cost, which can diminish their purchase intentions.

In addition to operating their own livestreams, e-tailers can choose to collaborate with influential third-party streamers. This approach enables consumers to experience the network effect, which arises from the social externalities generated by the streamer’s influence [7]. Typically, there are two modes of collaboration between e-tailers and streamers. One is the agency mode, in which the streamer offers an agency contract that consists of a commission fee and a slotting fee. For example, Jiaqi Li, a top-tier streamer, commonly charges a commission fee rate of around 20%, along with slotting fees up to USD 5 800 per brand. The other is the reselling mode, where the streamer offers a reselling contract and plays a role in product rebranding and selling. For instance, Dongfang Zhenxuan, renowned for its private-label products, offers the reselling contracts and has launched over 400 products, reaching a customer base of 30.86 million. These two modes differ from two perspectives. From the e-tailers’ perspective, the agency mode positions them as retailers, whereas the reselling mode positions them as product suppliers. From the consumers’ perspective, the reselling mode provides higher product value due to trust in the streamer’s brand [4].

Therefore, an e-tailer can choose among four modes: traditional, self-streaming, agency, and reselling. We present the real-world examples corresponding to different modes in Table I. In addition, the type of live-streaming contract offered by an influential streamer can significantly impact the e-tailer’s decision. However, this factor has been largely overlooked in the existing literature on live streaming channel selection. To address this gap, our first research question is as follows: Given a live-streaming contract offered by an influential streamer, should the e-tailer adopt live streaming, and if so, should it choose the self-streaming or third-party streaming?^{4,5,6,7}

To address this question, we develop a game-theoretic model that incorporates an e-tailer and an influential streamer. Our analysis reveals that the type of contract and consumers’ behavior significantly influence the e-tailer’s channel strategy. Interestingly, under an agency contract, a higher network effect may backfire. The e-tailer chooses to collaborate with the streamer only when the network effect is moderate and the consumer’s hassle cost is relatively low (Proposition 3). In contrast, given a reselling contract, the e-tailer can partner with the streamer even when the network effect is strong or weak (Proposition 5).

While existing studies have examined e-tailers’ choices of live streaming modes, they have largely overlooked the role of streamers in product pricing and contract provision [9], [10]. In practice, influential streamers possess substantial follower bases, which give them greater bargaining power when negotiating live streaming contracts [1]. Our study differs from prior work by considering the streamer’s initiative in offering a choice between agency and reselling contracts. The second question this paper seeks to address is as follows: how should a streamer offer live streaming contracts, and what are the equilibrium strategies for both the streamer and the e-tailer?

Interestingly, our findings reveal that the contracts offered by the streamer may not always align with its profit-maximizing objective. When the network effect is weak and the hassle cost is relatively low, the streamer may deliberately offer contracts that deviate from its profit-maximizing structure to encourage e-tailer collaboration (Proposition 6). Conversely, when the network effect is strong and the hassle cost remains low, the e-tailer will always prefer to collaborate with the streamer to maximize profits. This, in turn, allows the streamer to select the contract that best aligns with its profit objectives. Specifically, when the network effect is strong and the hassle cost is moderately low, the streamer tends to favor an agency contract to enhance marginal revenue. When the hassle cost is moderately high, it is still inclined to offer an agency contract, but to stimulate consumer demand and increase total revenue. We also examine the e-tailer’s channel choice strategy. When the hassle cost is prohibitively high, the e-tailer will choose the traditional strategy. Otherwise, it will choose between self-streaming and third-party streaming. As the hassle cost increases, self-streaming becomes a more appealing option for the e-tailer.

Under different live-streaming contracts and operational modes, e-tailers strategically adjust their pricing to manage demand across traditional and live-streaming channels. Additionally, when partnering with streamers, e-tailers often encounter a co-opetitive dynamic. A key question for both

⁴<https://www.christopherward.com/int/home> (accessed on November 2, 2025)

⁵<https://news.qq.com/rain/a/20251014A0403600> (accessed on November 2, 2025)

⁶<https://www.36kr.com/p/1985631263949829> (accessed on November 2, 2025)

⁷<https://www.jiemian.com/article/8163610.html> (accessed on November 2, 2025)

scholars and industry practitioners is how prices and demand fluctuate between traditional and live-streaming channels under varying channel modes.

Contrary to previous literature, which suggests that live streaming cannibalizes traditional channel demand, we find that under a reselling mode, the introduction of live streaming can actually boost demand in the traditional channel (Proposition 2). This effect depends on the commission fee rate and the strength of the network effect. Furthermore, live streaming channel prices are not always lower than those in traditional channels (Propositions 1 and 4). Finally, we find that employing live streaming does not guarantee higher total demand (Propositions 2 and 4).

We structure the remainder of this article as follows. Section II provides a review of the related literature and delineates the contributions of this study. In Section III, we develop an analytical model that incorporates key features specific to live streaming e-commerce, such as the network effect, consumer trust level, and hassle cost. Section IV presents the core analytical results, and Section V concludes this article.

II. LITERATURE REVIEW

Our article connects with three main areas of literature related to live streaming: 1) live streaming e-commerce; 2) multichannel selection; and 3) live streaming contracts. We also present a summary of key differences between our study and the relevant literature in Appendix B of the Supplementary Material.

The first stream relevant to our paper is live streaming e-commerce. Researchers mainly use empirical methods to investigate the reasons consumers engage with live streams and their purchasing decisions [3], [4], [6], [11], [12]. From the consumer side, studies indicate that hedonic motivations such as happiness [13], pleasure [14], trust [12], and engagement [15] drive purchases. Utilitarian motivations, such as high product fit [16], product quality [17], and lucky draw offers [18], also contribute to consumer purchasing decisions. Additionally, streamer qualities—like credibility [19], attractiveness [20], [21], and streamer expertise (e.g., knowledge, assortment depth, and selling ability)—positively influence consumer engagement in live streaming [11], [17], [22]. Researchers have also developed analytical models to explore interactions between sellers and streamers. For instance, Ji et al. [23] examine how sellers adjust their pricing and quality strategies when working with celebrity streamers. They find that sellers might lower the quality of products sold by celebrity agents. Li et al. [1] demonstrate that firms' strategies depend on the notoriety and bargaining power of a streamer. Meanwhile, Lin et al. [24] find that streamers often negotiate for deeper discounts, which can reduce profits for brands. Building upon the existing literature, our study considers key features of live streaming, such as the network effect, hassle cost, and consumer trust level, and examines their impact on pricing, demand, and profits. We also explore the interaction between an e-tailer and a streamer through the lens of contract selection. Using a game-theoretic approach, we analyze how the choice between self-streaming and third-party live streaming influences the e-tailer's channel selection. Whereas previous

studies focus on factors that influence live streaming participation, we aim to enhance cooperation between e-tailers and streamers by adopting different types of contracts.

The second stream closely related to this study is live streaming channel selection. Xu et al. [25] investigate retailers' live streaming strategies in response to manufacturer encroachment. Their results indicate that e-tailers may gain advantages from this encroachment, provided that live streaming's impact on the direct channel is substantial. Pan et al. [9] investigate the live streaming channel's implementation and conclude that the addition of live streaming is profitable only when the streamer possesses high selling ability. Zhang et al. [26] examine the vertical quality differences between products and discuss competition between platforms and manufacturers. Some studies have examined different formats of live streaming [27], [28], [29]. For example, Liu et al. [27] find that the live streaming advantages for e-tailers are contingent upon the slotting fee. Gong et al. [28] point out that product standardization and quality are crucial for e-tailers in adopting live streaming. They observe that low-quality products can lead to a cannibalization effect, reducing total market demand. Zhang et al. [29] find that retail prices depend on price intensity. Ma and Yang [30] consider a multinational firm using agency, resale, and live streaming channels and find that optimal channel selection depends on tax differences and channel substitutability. Some studies also connect e-tailers' channel selection with specific scenarios, including buy online and return in store [31], fresh industry [32], and AI streaming [33], [34]. These articles provide useful insights on the live streaming channel literature.

This study differs from prior research in several aspects. First, we build on studies [9] and [26] by incorporating the e-tailers' selection of various live streaming formats, including self-streaming and third-party live streaming. Second, our results reveal two important factors influencing e-tailers' channel selection: hassle cost and network effect. These factors lead to different results compared to previous literature. For example, in contrast to study [28], we find that the network effect may prevail over the cannibalization effect, leading to a net increase in total demand. Unlike study [27], we find that an e-tailer's profit could still suffer negative impacts even in the absence of a slotting fee. This occurs because consumer behavior influences channel competition and pricing decisions.

The third stream related to our paper concerns live streaming contracts. Existing literature on live streaming contracts predominantly addresses topics including compensation mechanism design, moral hazard, and sales commitment [35], [36], [37], [38], [39], [40]. For instance, in terms of compensation mechanism design, Zhang and Xu [35] compare proportional and linear incentive contracts, finding that both could incentivize streamers, but the former might have an advantage over the latter. Yang et al. [36] find that platforms can always benefit by providing bonuses to streamers. From a moral hazard perspective, Zhang et al. [38] find that a supplier can achieve greater benefits by coordinating with high-influence streamers who demand low price discounts. Zhang and Wang [37] employ the principal-agent model and uncovered that low signing bonuses could backfire on the platforms. Regarding sales commitments, Liu and Liu

[39] use a differential demand function to shed light on optimal decisions between a streamer and a platform committed to a revenue-sharing contract to incentivize product sales. They find that higher commission fee rates might not benefit the platform in deterring streamers participation. Xiao et al. [40] examine streamer sales commitments and seller choices, revealing that streamers prefer sales and effort commitments, whereas sellers prefer sales-only commitments. Some articles have focused on innovative live streaming contract modes, such as reselling or agency selling contracts [41], revenue sharing contract and flowed-based contract [10].

Our article is closely related to Wang et al. [10], who emphasize that the streamers' flow effect and promotion cost coefficient are crucial factors for manufacturers in determining whether to introduce the influencer live streaming and the associated contract types. Our work differs in three key respects: First, study [10] only focuses on the third-party live streaming format; however, our paper examines both e-tailer self-streaming and the third-party live streaming selling format. Second, study [10] introduce a novel flow-based contract that leverages streamer popularity to influence sales demand. Our study not only examines two innovative contract types offered by streamers but also explores the strategic interactions between streamers and e-tailers. Third, our work also addresses the streamer's contract offering strategy, aligning with the reality that large streamers have higher bargaining power, prompting e-tailers to seek collaboration with them.

This study contributes to the live streaming contracts literature in three distinct ways. First, we show that consumer behavior plays an important role in stakeholders' decision-making. Streamers can offer different contract types based on consumer characteristics, such as the network effect and hassle cost. Second, a streamer may choose contract types that do not align with their profit preferences to cooperate with e-tailers. Third, when a streamer provides an agency contract, a network effect that is either too strong or too weak may negatively affect the e-tailer.

III. MODEL

This study examines a live streaming scenario with two key players: an e-tailer that promotes products to end consumers through various distribution channels and an influential third-party streamer who presents two contract options to the e-tailer. Below, we detail our model construction.

A. E-Tailer

The e-tailer has three potential channel strategies, denoted as $X = \{T, S, C\}$. In the traditional strategy (referred to as "T"), the e-tailer initially sells the product directly to consumers through a traditional channel at a price p_t [9], [42]. Following the existing literature, we ignore the costs associated with the traditional channel, as they do not qualitatively affect our main results [9], [46].

With the rise of technologies such as live streaming, e-tailers have begun integrating live streaming into their sales strategies to expand market reach. One such approach is the self-streaming strategy (denoted as "S"), in which the e-tailer launches an

official live streaming channel to demonstrate and explain products directly to consumers. The e-tailer simultaneously sets the traditional channel price p_t and live streaming channel price p_l and retains full profits from both channels [1]. However, operating dual channels can intensify interchannel competition and incur additional operational costs in the live streaming channel. In the main analysis, we initially assume that the e-tailer incurs no operational cost c from self-streaming, which allows us to concentrate on the dual-channel competition. In Appendix C of the Supplementary Material, we relax this assumption by incorporating the operational cost c and find that our results remain robust.

Besides self-streaming, e-tailers may also partner with third-party streamers who have large follower bases. However, such collaborations often require e-tailers to share a portion of their profits and pricing power with streamers in exchange for better promotion. In addition, introducing a live streaming channel may also expose e-tailers to intensified price competition across sales channels and sacrifice their pricing power. This third-party strategy is denoted as C.

B. Third-Party Live Streamer

In practice, the streamer can offer two types of contracts, denoted as $Y = \{A, R\}$, where A represents the agency contract and R denotes the reselling contract. Under the agency contract, the streamer and e-tailer share aligned objectives and jointly divide the live streaming sales revenue. The streamer acts as an intermediary, selling products on behalf of the e-tailer. In contrast, under the reselling contract, the streamer gains greater autonomy over product selection and sales.

Specifically, under the agency contract, the streamer offers the e-tailer a two-part tariff contract $C(\phi, T)$ with a commission fee rate ϕ and a fixed slotting fee T . The streamer earns a proportion of the sales revenue and the slotting fee, while also retaining control over setting the live streaming channel p_l . Professional streamers, especially those with significant influence and audience trust, often possess greater bargaining power than e-tailers. This allows them to dictate pricing decisions to maximize sales revenue.⁸ This assumption is consistent with prior research, which highlights the streamer's dominant role in setting prices and contract terms [27], [29]. In the main analysis, we begin by assuming $T = 0$ to simplify the examination of the agency contract. This assumption aligns with current industry trends, where streamers often forgo slotting fees to attract e-tailers. For example, some large streamers have publicly announced their decisions to remove slotting fees [1]. In Appendix C of the Supplementary Material, we further explore the scenario in which the slotting fee is positive, (i.e., $T > 0$) to test the robustness of our model and examine its implications for both parties.

Under the reselling contract, the e-tailer acts as a supplier and sets the wholesale price r and traditional channel price p_t simultaneously. The streamer then customizes the packaging to

⁸<https://xueqiu.com/1080431220/265769127> (accessed on October 14, 2025)

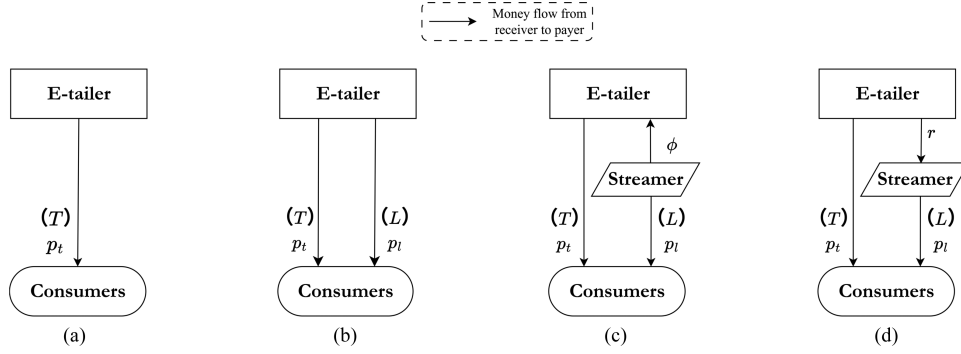


Fig. 1. E-tailer's channel modes. (a) Mode T. (b) Mode S. (c) Mode A. (d) Mode R.

TABLE II
NOTATION

	Symbols	Descriptions
Parameters	v	Consumer valuation, $v \sim U[0,1]$
	ϕ	Commission fee rate, $\phi \in (0,1)$
	γ	The strength of network effect, $\gamma \in (0,1)$
	$1 - \delta$	Uncertainty level, $\delta \in (0,1)$
	h	Consumer hassle cost, $h \in (0,1)$
	θ	Consumer trust level, $\theta \in (0,1)$
	c	Operational cost, $c > 0$
Functions	T	Slotting fee, $T > 0$
	U_i	Consumer utility of channel i
	D_i	Demands from channel format i
	D_i^k	Demand of channel i in mode k
	p_i^k	Price of channel i in mode k
	Π^k	Profits of the e-tailer in mode k
	π^Y	Profits of the live streamer under contract Y
Decision variables	p_i	Retail prices from channel format i
	r	Wholesale price
Others	i	Subscripts $i = \{t, l\}$ representing the traditional ($i = t$) and live streaming channel ($i = l$), respectively
	k	Superscript $k = \{T, S, A, R\}$ representing traditional mode ($k = T$), self-streaming mode ($k = S$), agency mode ($k = A$) and reselling mode ($k = R$), respectively
	X	$X = \{T, S, C\}$ representing e-tailer traditional strategy ($X = T$), self-streaming strategy ($X = S$) and collaboration with a third-party streaming strategy ($X = C$), respectively.
	Y	Superscript $Y = \{A, R\}$ representing agency contract ($Y = A$), reselling contract ($Y = R$), respectively.

rebrand the products as its own, selling them in the live streaming channel at a retail price p_l [42].

Accordingly, we define four distinct modes based on the e-tailer's sales strategy and contractual relationship with the streamer. In the traditional mode, the e-tailer sells products exclusively through its traditional channel. In the self-streaming mode, the e-tailer simultaneously operates both the traditional and its own live streaming channels. In both the agency mode and the reselling mode, the e-tailer offers products through the traditional channel as well as via a third-party streamer engaged through either an agency or a reselling contract, respectively. Fig. 1 illustrates the e-tailer's choices, and Table II summarizes the notation used throughout the paper.

C. Consumer Choice Model

In this section, we examine the consumer choice model under varying e-tailer strategies. Consumers exhibit heterogeneous

valuations for the product, represented by v , which is uniformly distributed over the interval $[0, 1]$ [46]. We define U as a consumer's utility.

In the traditional mode, consumers face limited information regarding product attributes such as content, quality, and fit, which leads to product uncertainty [6], [34], [43]. As a result, consumers form a valuation of the product as δv , where $1 - \delta$ captures the uncertainty level and $0 < \delta < 1$. Therefore, the utility for a consumer purchasing through the traditional channel is $U_t = \delta v - p_t$.

With the introduction of e-tailer's self-streaming, consumers can choose between the traditional and live streaming channels for their purchases. Through real-time product demonstrations and explanations, live streaming enables consumers to reduce product valuation uncertainty. However, consumers incur a hassle cost h ($0 < h < 1$) when engaging with live streaming [28], [43]. For instance, consumers may need to wait for the streamer's explanation rather than being able to purchase immediately, and

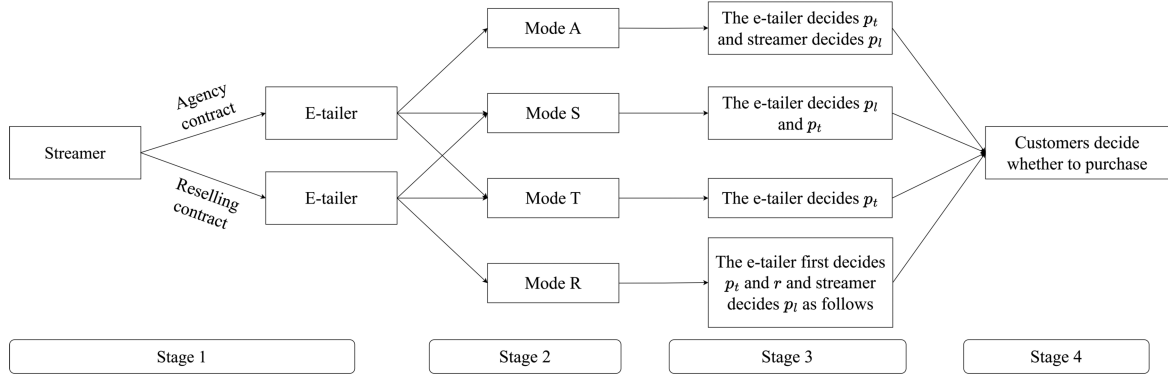


Fig. 2. Sequence of events.

their questions may not be addressed in a timely manner, which can negatively affect the consumer experience [9]. Accordingly, the utility derived by a consumer from the live streaming channel under mode S is expressed as $U_l = v - h - p_l$. In the main analysis, we initially assume that the strength of the network effect is zero. The rationale for this assumption is that the strength of the network effect is much weaker compared to the third-party streaming. In Appendix C of the Supplementary Material, we relax this assumption by incorporating the positive strength of network effect and find that our findings remain robust.

When the e-tailer engages a third-party streamer, live streaming not only reduces consumers' uncertainty about product quality but also introduces a network effect that enhances their experience. The network effect refers to the social externalities generated within a live-streaming room, whereby a consumer's utility increases as more consumers purchase the same product. This incremental utility is represented as γD_l . Here, γ represents the strength of the network effect, reflecting the streamer's influence. D_l represents the live streaming demand. We assume consumers have rational expectations, meaning that their expected number of purchasers equals the actual number. Furthermore, we assume γ is bounded above by a certain threshold (i.e., $\gamma < \min\{1 - \delta, h\}$). This assumption reflects that consumers primarily rely on their intrinsic preferences when making purchase decisions, with social influence serving as a secondary factor. This assumption also ensures the profit function is concave and that an equilibrium exists.

It is worth noting that, under mode R, the streamer sells private-label products, and consumers' trust in the streamer's brand positively influences their valuation [20]. We define this consumer trust effect as θv , where θ represents the consumer trust level. We also assume that θ is bounded above by a certain threshold (i.e., $\theta < \delta$), reflecting that consumer trust in the streamer's selection is limited. For example, Dongfang Zhenxuan, an MCN that procures goods through a rigorous curation process and sells them under its brand during livestream sessions.⁹ Accordingly, the consumer derives a utility when purchasing through the live streaming channel as $U_l = v + \gamma D_l - h - p_l$ under mode A, and as $U_l = (1 + \theta)v + \gamma D_l - h - p_l$ under mode R.

⁹<https://news.qq.com/rain/a/20221013A06WD900> (accessed on August 28, 2025)

The game sequence is illustrated in Fig. 2. In Stage 1, the streamer determines which type of contract to offer. Upon observing the streamer's contract decision, the e-tailer then decides whether to engage in collaborating with the streamer in Stage 2. If the e-tailer chooses not to collaborate, it can either self-streaming or traditional selling. The e-tailer makes pricing decisions in Stage 3, while the streamer (if involved) simultaneously determines its own price. Finally, in Stage 4, customers decide whether to make a purchase and, if so, through which channel.

IV. MAIN RESULT

This section presents the primary findings of this study. First, we focus on the subgame perfect Nash equilibrium when streamer offers the agency contract and reselling contract, respectively. We then move to the first stage and derive the streamer's equilibrium contract offering strategies and e-tailer's channel selection strategies. The subgame optimal solutions under each mode are summarized in Table A2 of Appendices A and B of the Supplementary Material, while the technical derivations are provided in Appendix D of the Supplementary Material. The structural properties of each mode are presented in Appendix B of the Supplementary Material.

A. When the Streamer Offers an Agency Contract

When the streamer opts to offer an agency contract, the e-tailer selects a channel strategy from the set $X = \{T, S, C\}$.

1) *Not Adopting Live Streaming*: When the e-tailer decides not to adopt live streaming (Mode T), consumers make purchases solely from the traditional channel. The consumer's utility function is defined as $U_t = \delta v - p_t$. The marginal consumer who is located at $v_t = \frac{p_t}{\delta}$ is indifferent between making a purchase or not. Consequently, the demand for traditional channel is $D_t = 1 - v_t$ while the e-tailer's profit function is $\Pi = p_t D_t$. In this mode, the e-tailer sets the traditional channel p_t .

2) *Introducing Self-Streaming*: When the e-tailer chooses to self-stream, consumers can select either the traditional channel or the live streaming channel to purchase. The utility of purchasing through the traditional and live streaming channels is $U_t = \delta v - p_t$ and $U_l = v - p_l - h$, respectively. The consumers who is indifferent between purchasing through the two channels is located at $v_{st} = \frac{h + p_l - p_t}{1 - \delta}$. The demands for the two

channels are $D_l = 1 - v_{st}$ and $D_t = v_{st} - v_t$, respectively. The e-tailer's profit function is $\Pi = p_t D_t + p_l D_l$ and it sets p_l and p_t simultaneously.

3) *Introducing Third-Party Streaming*: Given an agency contract, the e-tailer can also choose to accept this contract and collaborate with a third-party live streamer. The utility of purchasing through the traditional and live streaming channels is $U_t = \delta v - p_t$ and $U_l = v + \gamma D_l - p_l - h$, respectively. The marginal consumer who is indifferent between the two channels is located at $v_{at} = \frac{h + p_l - p_t - \gamma D_l}{1 - \delta}$. The demand functions are $D_t = v_{at} - v_t$ for the traditional channel and $D_l = 1 - v_{at}$ for the live-streaming channel. The profit functions of the e-tailer and the third-party live streamer are $\Pi = p_t D_t + (1 - \phi)p_l D_l$ and $\pi = \phi p_l D_l$, respectively. In this mode, the e-tailer sets p_t and streamer sets p_l simultaneously.

Proposition 1: When the live streamer offers the agency contract, the price comparisons are summarized as follows.

- 1) (a) $p_t^A < p_t^T = p_t^S$; (b) $p_l^A < p_l^S$.
- 2) (a) $p_t^S < p_t^T$; (b) if $h > h_{al}$ and $\gamma < \gamma_{al}$, $p_l^A < p_l^S$; otherwise, $p_l^A \geq p_l^S$. The expressions of h_{al} and γ_{al} can be found in Table A1 of Appendix A of the Supplementary Material.

Part 1) of Proposition 1 compares the pricing strategies across traditional and live streaming channels under different modes. First, the agency mode leads to a lower traditional channel price, whereas the self-streaming mode does not. Second, the live streaming channel price under the agency mode is lower than that under the self-streaming mode. These results are explained as follows.

In mode S, the e-tailer sets prices for both channels and engages in inter-channel price discrimination to target consumers. Since live streaming does not alter the consumer valuation in the traditional channel, the e-tailer has no incentive to adjust the traditional channel price. In contrast, the agency mode introduces a key shift: the streamer, not the e-tailer, determines pricing in the live streaming channel. This division of pricing authority creates intensified price competition for the e-tailer, which faces not only demand cannibalization from live streaming but also commission expenses paid to streamers. To mitigate these effects, the e-tailer is incentivized to lower the traditional channel price. This finding contrasts with [47], which suggests that the introduction of a live streaming channel does not affect the traditional channel price. The rationale for part (b) is that in mode A, the streamer has a stronger incentive to set lower prices to amplify the network effect, leading to a lower live streaming channel price compared with mode S. Part 2) examines the price comparison between dual channels. First, in mode S, the live streaming channel's price is higher than that of the traditional channel's because self-streaming mitigates consumer valuation uncertainty, allowing the e-tailer to charge more. The managerial implication of $p_t^S < p_t^T$ is that the e-tailer's live streaming channel tends to earn profits through higher prices rather than higher demand. This strategy helps protect the traditional channel's sales by avoiding direct price competition between the two channels. Second, in mode A, the price of the live streaming channel can be either higher or lower than that of the traditional channel. Only when the hassle cost is high and the network effect is weak does the

streamer need to lower the live streaming price to compensate for the inconvenience. When the hassle cost is low, the appeal of live streaming decreases consumers' price sensitivity, allowing the streamer to charge a higher price. Conversely, when the hassle cost is high and the network effect is strong, increased competition results in lower prices in the traditional channel.

Our findings extend the existing literature on live streaming channels. For example, Li et al. [11] show that the live streaming price tends to be higher when selling products with low price sensitivity. Our study complements this literature by showing that consumer behavior and socialization also shape pricing strategies. In addition, Pan et al. [9] provide empirical evidence that supports the notion that the live streaming channel prices may be even higher, reinforcing our theoretical insights. From a managerial perspective, our results suggest that streamers can set a higher price by reducing consumers' hassle costs—e.g., providing timely responses—or by enhancing the network effect—e.g., encouraging interaction among consumers.

Proposition 2: When the live streamer offers the agency contract, the demand comparisons are summarized as follows.

- 1) (a) $D_t^S < D_t^T$; (b) $D_l^S < D_l^A$; (c) if $\phi > \phi_{as}$ and $\gamma < \gamma_{as}$, $D_t^T < D_t^A$; otherwise, $D_t^T \geq D_t^A$.
- 2) $D_t^S + D_l^S = D_t^T < D_t^A + D_l^A$. The expressions of ϕ_{as} and γ_{as} can be found in Table A1 of Appendix A of the Supplementary Material.

Part 1) shows that the e-tailer's self-streaming reduces demand in the traditional channel. This occurs because introducing a live streaming channel diverts some high-valuation consumers away from the traditional channel, thereby lowering the demand. Moreover, the live streaming channel demand under mode S is lower than that under mode A. This is because the streamer strategically lowers the price, which in turn boosts live streaming channel demand. Interestingly, under conditions of a high commission fee rate and weak network effect, the agency mode may even increase demand in the traditional channel. This counterintuitive outcome is explained as follows.

In mode A, a higher commission fee rate incentivizes the e-tailer to reduce the traditional channel price to attract more consumers. At the same time, when the network effect is weak, live streaming channel prices remain relatively high, further encouraging purchases in the traditional channel. Under these conditions, the additional demand generated by the traditional channel outweighs the cannibalization effect of the live streaming channel, resulting in $D_t^T < D_t^A$. Conversely, when the commission fee rate is high and the network effect is strong, live streaming becomes more appealing, diminishing consumer motivation to purchase through the traditional channel. On the other hand, when the commission fee rate is low, the e-tailer raises the price in the traditional channel, intensifying demand cannibalization and resulting in a lower traditional channel demand.

Part 2) shows that under mode A, introducing third-party live streaming increases the e-tailer's total demand, whereas self-streaming does not. This occurs because the agency mode amplifies inter-channel competition. To retain customers and mitigate live streaming's cannibalization, the e-tailer lowers the traditional channel price to attract low-valuation consumers,

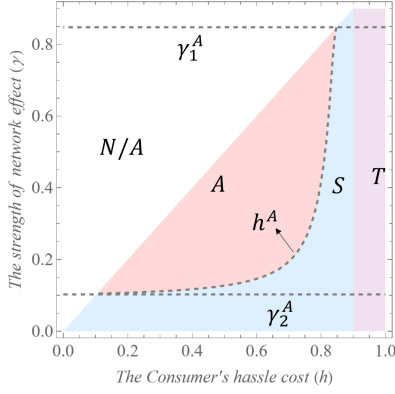


Fig. 3. Channel selection under agency contract (Parameters: $\delta = 0.1$, $\phi = 0.1$).

expanding traditional channel demand and resulting in a market expansion effect. This mechanism explains why some e-tailers collaborate with third-party streamers to boost overall sales in practice.

The managerial implications of Proposition 2 are outlined as follows. First, e-tailers should recognize that introducing a live streaming channel does not necessarily result in demand cannibalization. Second, the traditional channel demand is more susceptible to cannibalization under the self-streaming mode than under the agency mode. Third, e-tailers aiming to maximize sales should consider partnering with third-party streamers and adopt the agency mode.

Proposition 3: When the live streamer offers the agency contract, the Subgame Perfect Nash Equilibrium is summarized as follows.

- 1) If $h > 1 - \delta$, the e-tailer will choose mode T.
- 2) If $\gamma_1^A < \gamma < \gamma_2^A$ and $h < h^A$, the e-tailer will choose mode A.
- 3) Otherwise, the e-tailer will choose mode S. The expressions of γ_1^A , γ_2^A and h^A can be found in Table A1 of Appendix A of the Supplementary Material.

Proposition 3 states that when the live streamer offers the agency contract to the e-tailer, if the hassle cost is high, the e-tailer will choose not to adopt live streaming. If the network effect is moderate and the hassle cost is low, the e-tailer will accept the third-party live streaming mode. In all other cases, the e-tailer will opt for the self-streaming mode. Fig. 3 illustrates Proposition 3, and the underlying rationale is explained as follows.

First, when the hassle cost is high (i.e., $h > 1 - \delta$), it outweighs benefits of live streaming, thereby dampening consumer surplus in live streaming channel. Consequently, both the e-tailer and the live streamer must cut live streaming channel prices, leading to a reduction in profit margins. To avoid such profit loss, the e-tailer chooses not to adopt live streaming and relies solely on the traditional channel. Conversely, when the hassle cost is low, the value consumers gain from reduced uncertainty exceeds hassle cost-induced disutility, making live streaming a viable strategy. When both the network effect and the hassle cost is weak (i.e., $\gamma_1^A < \gamma < \gamma_2^A$ and $h < h^A$), the third-party live streaming fails to enhance consumer engagement, leading

to relatively low channel demand. Consequently, the benefits of third-party streaming are insufficient to offset losses from channel competition and commissions, making self-streaming more profitable. When the network effect is strong, both the traditional and live streaming channels experience lower marginal revenues under the agency mode. Despite an overall increase in market demand, the reduced marginal revenue fail to compensate for the potential losses. Under these market conditions, self-streaming remains the more advantageous option for the e-tailer.

Proposition 3 also yields an insightful observation. In contrast to prior studies that suggest network effect tend to enhance the e-tailer's sales advantage [48], our findings reveal that such social influence can intensify interchannel competition, ultimately eroding the e-tailer's profit. This implies that e-tailers may benefit more from partnering with mid-tier streamers rather than top-tier streamers given the agency contract. Second, as hassle cost increase, the e-tailer's channel strategy shifts from an agency mode to a self-streaming mode. This insight highlights the importance for e-tailers to align their live streaming strategies with prevailing market conditions. Third, introducing self-streaming may lead to higher profitability, a conclusion that aligns with the observed trend toward increased adoption of self-streaming.

B. When the Streamer Offers the Reselling Contract

When the streamer offers a reselling contract, the e-tailer selects a live streaming strategy from the set $X = \{T, S, C\}$. Strategy T and S remain the same in Section IV-A, we omit the detailed discussion here. In this subsection, we examine the scenario in which the e-tailer chooses to partner with a third-party live streaming ($X = C$), under the reselling contract.

When the e-tailer selects a third-party streamer to sell products under a reselling mode, the utility functions for consumers purchasing through the live streaming channel and the traditional channel are given by $U_l = (1 + \theta)v + \gamma D_l - h - p_l$ and $U_t = \delta v - p_t$, respectively. The marginal consumer indifferent between the two channels is located at $v_{rt} = \frac{h + p_l - p_t - \gamma D_l}{1 - \delta + \theta}$. The demand for the live streaming channel is $D_l = 1 - v_{rt}$ and the demand for the traditional channel is $D_t = v_{rt} - v_t$. The profit functions of the e-tailer and the streamer are defined as $\Pi = p_t D_t + r D_l$ and $\pi = (p_l - r) D_l$, respectively. In this mode, the e-tailer simultaneously determines the wholesale price r and the traditional channel price p_t , whereas the streamer determines the price p_l .

Proposition 4: When the live streamer offers the reselling contract, the price comparisons are summarized as follows.

- 1) (a) $p_t^S = p_t^R = p_l^T$; (b) $p_l^S < p_l^R$; (c) $p_t^R < p_l^R$.
- 2) (a) $D_l^R < D_t^R < D_t^T$; (b) If $\gamma < \frac{1-\delta}{2}$, $h > \frac{1-\delta}{2}$ and $\theta < \max\{\theta_{RS}, \delta\}$, $D_l^R < D_l^S$, otherwise, $D_l^R \geq D_l^S$; (c) $D_t^S + D_l^S = D_t^T = D_t^R + D_l^R$. The expression of θ_{RS} can be found in Table A1 of Appendix A in the Supplementary Material.

Part 1) examines how the introduction of a reselling contract by the streamer affects the e-tailer's pricing decisions across different modes. Our findings indicate that traditional

channel price remains stable under modes S and R. However, the mechanisms underlying these scenarios differ. When the e-tailer engages in self-streaming, it maintains control over pricing in both channels. To maximize overall profit, it sets a relatively higher price in the live streaming channel while keeping the traditional channel price unchanged. Conversely, in mode R, the e-tailer and the streamer share revenue in the live streaming channel while competing on price in the traditional channel. This creates a co-opetitive relationship between the e-tailer and the streamer. To reduce price competition and protect the profitability of the traditional channel, the e-tailer strategically increases the wholesale price. As a result, the traditional channel remains stable.

Part 1) (b) and (c) indicate that under the reselling mode, prices in the live streaming channel surpass those in the traditional and self-streaming modes. We explain this phenomenon from two perspectives. First, under the reselling mode, consumers are influenced by their trust in the streamer's brand, which enhances their perceived utility and reduces their price sensitivity. This feature enables streamers to set higher prices within the live streaming channel. Second, the reselling mode is subject to the double marginalization effect: since the e-tailer charges a wholesale price, the streamer has an incentive to set higher prices to preserve its profit margin. This finding aligns with real-world observations where products sold directly by streamers or MCN organizations often carry a price premium compared to their suppliers' retail channels. For instance, Dongfang Zhenxuan sells corn at approximately USD 0.83 per piece, whereas its supplier, "Northeast Farm Aunt," offers the same product for only USD 0.5.¹⁰

Part 2) examines the relationship between dual-channel demands across different sales modes. First, under the reselling mode, the demand in the live streaming channel is lower than that in the traditional channel. In addition, the traditional channel demand under mode T exceeds both channel demands under mode R. This is because only consumers with high valuations are willing to purchase through the live streaming channel, resulting in lower demand relative to the traditional channel. Also, because of the inter-channel competition, introducing a live streaming channel encroaches on traditional channel demand, leading to a reduction in traditional channel demand compared with mode T. Second, the total market demand remains unchanged under the reselling mode. The underlying reason is that the e-tailer does not need to lower the traditional channel's price to attract low-valuation consumers, leaving total demand unaffected. Third, it is noteworthy that the demand for the live streaming channel under the reselling mode is not necessarily higher than that under the self-streaming mode. This outcome arises when the consumer trust level is relatively weak. In such cases, if the hassle cost associated with the live streaming channel is high, consumers may be less inclined to use it, thereby reducing live-streaming demand under the reselling mode compared to the self-streaming mode.

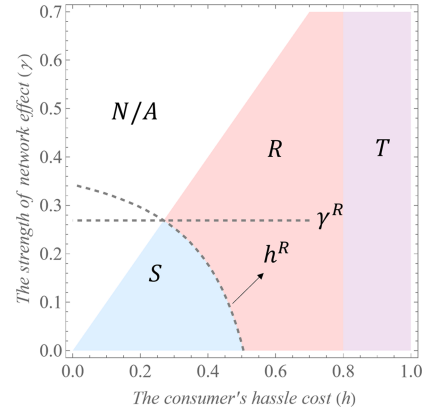


Fig. 4. Channel selection under reselling contract (Parameters: $\delta = 0.3, \theta = 0.1$).

Proposition 5: When the live streamer offers the reselling contract, the subgame perfect Nash equilibrium is given as follows.

- 1) If $h > 1 - \delta + \theta$, the e-tailer will choose mode T.
- 2) If $\max\{\gamma, h^R\} < h < 1 - \delta + \theta$, the e-tailer will choose mode R.
- 3) Otherwise, if $\gamma < \gamma^R$ and $\gamma < h < h^R$, the e-tailer will choose mode S. The expressions of γ^R and h^R can be found in Table A1 of Appendix A of the Supplementary Material.

Proposition 5 analyzes the e-tailer's strategic choices under the reselling contract offered by the live streamer. As shown in Fig. 4, the e-tailer will not adopt the live streaming mode when the hassle cost h is high (i.e., $h > 1 - \delta + \theta$). This is because a high hassle cost leads to profit erosion from the live streaming channel discounts, making it unappealing. Second, when the hassle cost h is moderate (i.e., $\max\{\gamma, h^R\} < h < 1 - \delta + \theta$), the e-tailer will choose to collaborate with the third-party live streamer. This is because, on one hand, the hassle cost reduces consumer interest in self-streaming, thereby diminishing the e-tailer's marginal benefit from this mode. On the other hand, consumers' trust in the streamer's products allows the e-tailer to set a higher price. This price premium effectively offsets potential profit losses associated with hassle cost, making collaboration more profitable than self-streaming. Third, when both the strength of the network effect and the hassle cost are low (i.e., $\gamma < \gamma^R$ and $\gamma < h < h^R$), the e-tailer prefers to engage in self-streaming rather than collaborate with the live streamer. This is because a weak network effect generates limited additional demand, diminishing the relative advantage of the agency mode.

Comparing Proposition 5 with Proposition 3, we derive an important finding: unlike the agency contract, the reselling contract can make cooperation with the streamer attractive to the e-tailer even when the hassle cost is high. In contrast, under the agency mode, high hassle cost pushes the e-tailer to opt for self-streaming. This divergence arises because the reselling mode offers two key advantages to the e-tailer. First, the e-tailer profits from the live streaming channel by setting the wholesale price, which mitigates channel competition. Second, consumers

¹⁰<https://www.36kr.com/p/1936212472629633> (accessed on August 28, 2025)

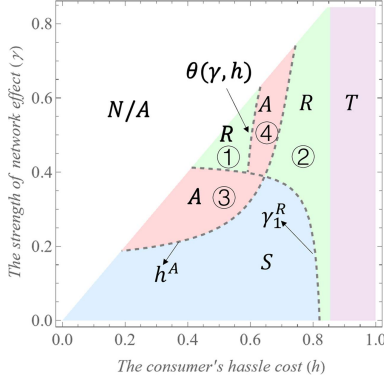


Fig. 5. Equilibrium mode selection (Parameters: $\delta = 0.156$, $\theta = 0.01$, $\phi = 0.17$).

are willing to pay more for the additional value generated by live streaming. Consequently, the increase in marginal revenue offsets the potential costs generated by hassle cost, making collaboration with the streamer attractive to the e-tailer even when the hassle cost is high.

Proposition 5 provides several important managerial implications for e-tailers. Specifically, e-tailers should carefully evaluate how different contractual arrangements offered by streamers influence channel choices. For instance, under both weak and strong network effect, partnering with a streamer may still be advantageous if the streamer adopts a reselling contract. However, the ultimate desirability of such a collaboration hinges on the magnitude of the hassle cost.

C. Equilibrium Strategies of the Live Streamer and the E-Tailer

In this section, we summarize the streamer's contract choice in Stage 1 to identify both the streamer's equilibrium contract offering strategy and the e-tailer's channel selection strategy. Technical details are provided in Appendix D of the Supplementary Material.

Proposition 6: Equilibrium strategies for streamer contract offering and e-tailer's channel selection are given as follows.

- 1) If (a) $\gamma > \gamma_1^{R2}$, $\gamma < h < h_{AD}^A$ and $\theta < \theta(\gamma, h)$ (b) $\gamma < \gamma_1^R$ and $\gamma < h < h_{AD}^A$, the streamer offers the agency selling contract and the e-tailer accepts it.
- 2) If (a) $\gamma > \gamma_1^{R2}$, $\gamma < h < h_{AD}^A$ and $\theta > \theta(\gamma, h)$ (b) $\gamma < \gamma_1^R$ and $h > h_{AD}^A$ (c) $1 - \delta < h < 1 - \delta + \theta$, the streamer offers the reselling contract and the e-tailer accepts it.
- 3) Otherwise, the e-tailer will not cooperate with the streamer. Specifically, mode T is chosen if $h > 1 - \delta + \theta$. Mode S is preferred if $0 < \gamma < \gamma_1^R$ and $h_{AD}^A < h < 1 - \delta$. The expressions of $\theta(\gamma, h)$, γ_1^{R2} , h_{AD}^A , γ_1^R , and h_{AD}^A are listed in Table A1 of Appendix A of the Supplementary Material.

Proposition 6 sheds light on the interplay between the streamer's contract choices and e-tailer's channel strategies. The streamer offers a reselling contract when the network effect is strong and hassle cost is moderately low (Region ① in Fig. 5), or when both the network effect and the hassle cost are relatively high (Region ②). Conversely, it offers an agency contract when

the network effect is strong and hassle cost is moderately high (Region ④) or when both are relatively low (Region ③). In the aforementioned scenarios, the e-tailer prefers cooperation with the streamer. However, when the network effect is weak and the hassle cost is moderately high, the e-tailer opts for self-streaming. In cases where the hassle cost is prohibitively high, the e-tailer reverts to the traditional mode. We provide an explanation of Proposition 6 as follows.

Part 1) examines the streamer's contract offering strategy. When deciding which contract to offer, the streamer weighs its profit preferences against the e-tailer's contract acceptance incentives. This strategic interaction results in two types of contract offerings: the optimal offering strategy and the suboptimal offering strategy. As illustrated in Fig. 5, regions ① and ④ represent scenarios in which the streamer adopts an optimal strategy. Specifically, when the network effect is strong and the hassle cost is low—recall Propositions 3 and 4—the e-tailer prefers partnering with the streamer. This situation allows the streamer to choose the contract type that maximizes its profit.

As shown in region ①, the streamer prefers to offer a reselling contract when the hassle cost is low. This is mainly because a lower hassle cost allows streamers to command higher prices under a reselling contract, yielding greater profit. Conversely, high hassle costs constrain pricing, leading streamers to prioritize demand via live streaming and offer an agency contract (region ④). This finding aligns with real-world practices: influential streamers such as Jiaqi Li often provide agency contracts to e-tailers to drive the demand, whereas some streamers who sell their own branded products, like Dongfang Zhenxuan, prefer reselling contracts to benefit from elevated price.

It is interesting to highlight that regions ② and ③ highlight suboptimal strategies, where the streamer's profit preference may not align with its contract offering. This misalignment arises because the streamer aims to maximize the likelihood of securing a partnership to expand its market coverage. When both the network effect and hassle cost are low (see Proposition 3), partnering with the streamer is more advantageous for the e-tailer compared to self-streaming or traditional selling. To generate positive profits, the streamer offers an agency contract to entice e-tailer collaboration. Similarly, when the network effect is strong and the hassle cost is either moderately high or very high (see Proposition 4), the e-tailer is more inclined to collaborate under a reselling contract. As a result, the streamer chooses to offer a reselling contract even if doing so deviates from its profit-maximizing preference. Otherwise, the streamer earns zero profit.

Part 2) analyzes the e-tailer's channel provision strategy. When the network effect is weak and the hassle cost is moderately high, consumers find third-party live streaming less appealing. In this situation, if the e-tailer partners with a streamer, it faces revenue-sharing losses and encounters limited demand through the live streaming channel, thereby reducing overall profitability. As a result, the e-tailer may choose to adopt a self-streaming mode. When the hassle cost is significant, live streaming prices under different modes are low. To avoid profit erosion, the e-tailer relies solely on mode T. Alternatively, the e-tailer may opt to collaborate with the streamer and establish

contractual arrangements based on the market conditions discussed in Part 1).

The managerial implications of Proposition 6 are threefold. First, a stronger network effect not only increases the likelihood of collaboration with e-tailers but also provides streamers with greater bargaining power and flexibility in contract offering choice. This enables streamers to consider their own profit preferences rather than being solely subject to the e-tailer's terms. Streamers can strengthen the network effect within their live streaming sessions by leveraging personal charisma, curating engaging content, or offering promotional incentives to attract a larger audience. Second, e-tailers can improve the effectiveness of self-streaming livestreams by optimizing the sequence of product displays or by pre-releasing product information to reduce consumers' hassle cost. These strategies make self-streaming more competitive. Third, both streamers and e-tailers should pay close attention to consumer behaviors—particularly the strength of the network effect and hassle cost—as these factors critically inform decisions regarding channel selection and contract design. From a policy perspective, policymakers can foster policies that promote technological innovations such as AI-powered chatbots. These tools can reduce hassle costs and improve the consumer experience, thereby fostering the growth of live-streaming e-commerce.

V. CONCLUSION

As live streaming rapidly reshapes online retailing, understanding how streamers strategically design and offer contracts has become a critical yet underexplored issue. While prior research has primarily viewed live streaming as a sales or marketing channel, our study shifts the focus to the contractual interaction between third-party streamers and e-tailers, revealing how incentive design fundamentally shapes channel structure and market outcomes. Using a game-theoretic model, we analyze two types of contracts—agency and reselling—across three distinct channel strategies: third-party streaming, self-streaming, and not adopting live streaming. The main findings and managerial implications are presented below.

First, we find that streamers do not always select contracts that align with their short-run profit preferences. For example, when both the network effect and the hassle cost are low, a streamer chooses an agency contract rather than the reselling contract that would maximize its profit. Conversely, when the network effect is strong and the hassle cost is moderately high or very high, the streamer opts for reselling even if this deviates from its profit-maximizing preference. This misalignment reflects a deliberate trade-off between immediate profits and the stability of cooperation with the e-tailer. To prevent e-tailer exit, the streamer strategically offers the sub-optimal contract that secures participation and sustains the relationship.

Moreover, our results explain why both agency and reselling contracts coexist in practice, and clarify the underlying motivations behind streamers' choices. When the network effect is strong and the hassle cost is moderate to low, a streamer prefers to offer a reselling contract in order to capture higher margins through increased price. Conversely, when the network

effect remains strong but the hassle cost is moderate to high, the streamer tends to adopt an agency contract to benefit from the expanding live streaming demand. This contrast reflects the streamer's strategic tradeoff between winning by price and winning by volume under different market conditions. In practice, for example, Jiaqi Li often chooses agency contracts to share in demand surges, whereas Dongfang Zhenxuan tends to adopt reselling contracts to raise selling prices and expand profit.

Second, no single strategy is universally dominant in equilibrium. This finding aligns with real-world practices, where different e-tailers adopt distinct live streaming modes. For instance, Apple and Peacebird relies on self-streaming to maintain strict control over pricing, while brands such as Fresh and Lancôme collaborate with streamers to leverage network effect within the live streaming environment. In contrast, brands like Christopher Ward, almost entirely refrain from live streaming. Our finding suggest that e-tailers should consider two critical factors when formulating channel strategies: the strength of network effect and the consumer's hassle cost.

Third, unlike prior studies that suggest network effects tend to enhance the e-tailer's sales advantage, our findings reveal that when the streamer adopts an agency contract, stronger network effects can intensify interchannel competition, ultimately eroding the e-tailer's profits under the agency mode. This insight helps explain why, in practice, e-tailers often prefer to collaborate with mid-tier streamers rather than top-tier influencers when agency contracts are in place. The underlying reason is mid-tier streamers can efficiently reach target audiences without triggering excessive inter-channel competition or commission pressure. This also sheds light on why platforms such as Taobao Live and Douyin have introduced policies to support mid-tier streamers, aiming to sustain a more balanced and cooperative ecosystem between e-tailers and streamers.

Finally, while conventional studies suggest that live streaming attracts consumers mainly through lower prices, our analysis shows that this is not always the case under the reselling mode for consumers are willing to pay a trust premium. This aligns with the reality where Dongfang Zhenxuan has been reported to charge higher prices in its live streams. Moreover, live streaming does not inevitably cannibalize traditional channels or expand overall demand; these outcomes hinge on core features such as the live streaming mode, the commission fee rate, and the strength of network effect. We therefore recommend that e-tailers and streamers closely assess market conditions to set prices appropriately and to better anticipate demand shifts.

This study has several limitations that open avenues for future research. First, this study analyzes the interaction between an e-tailer and a single live-streaming streamer. In practice, however, a streamer may collaborate with multiple e-tailers. This situation raises an intriguing question: how should a streamer design and offer contracts to competing e-tailers, and what contractual forms are optimal in such scenarios? Second, our current framework does not consider the e-tailer's strategic choice when third-party live streaming and self-streaming coexist. This omission is deliberate, as third-party influencer marketing is typically a short-term strategy aimed at rapid sales promotion, whereas self-streaming represents a long-term investment in brand building

and customer relationship management. Future research could consider how e-tailers should dynamically allocate resources between third-party and self-streaming over time. Such inquiries would contribute to a comprehensive understanding of live streaming-based commerce.

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